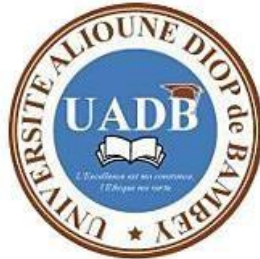


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Un peuple Un But Une Foi

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UNIVERSITE ALIOUNE DIOP DE BAMBEY



UFR : SCIENCES APPLIQUÉES ET TECHNOLOGIES DE L'INFORMATION ET DE LA COMMUNICATION(SATIC)

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NIVEAU : MASTER 1

Création d'une application Web dans un conteneur Docker

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Introduction

Ce rapport présente la réalisation complète du TP portant sur la création d'une application Web déployée dans un conteneur Docker. L'objectif principal est de comprendre le cycle complet de développement : écriture d'un script Bash, développement d'une application Python/Flask, puis containerisation avec Docker.

Les étapes réalisées sont :

- Création d'un script Bash interactif
- Développement d'une application Web avec Flask (Python)
- Configuration des fichiers HTML/CSS de l'application
- Création d'un script Bash d'automatisation Docker
- Construction et exécution du conteneur Docker

Partie 2 : Création d'un script Bash simple

Cette partie introduit les bases du scripting Bash, indispensables pour automatiser des tâches dans les environnements DevOps et CI/CD.

Étapes réalisées

Étape 1-3 : Création du fichier et ajout du she-bang

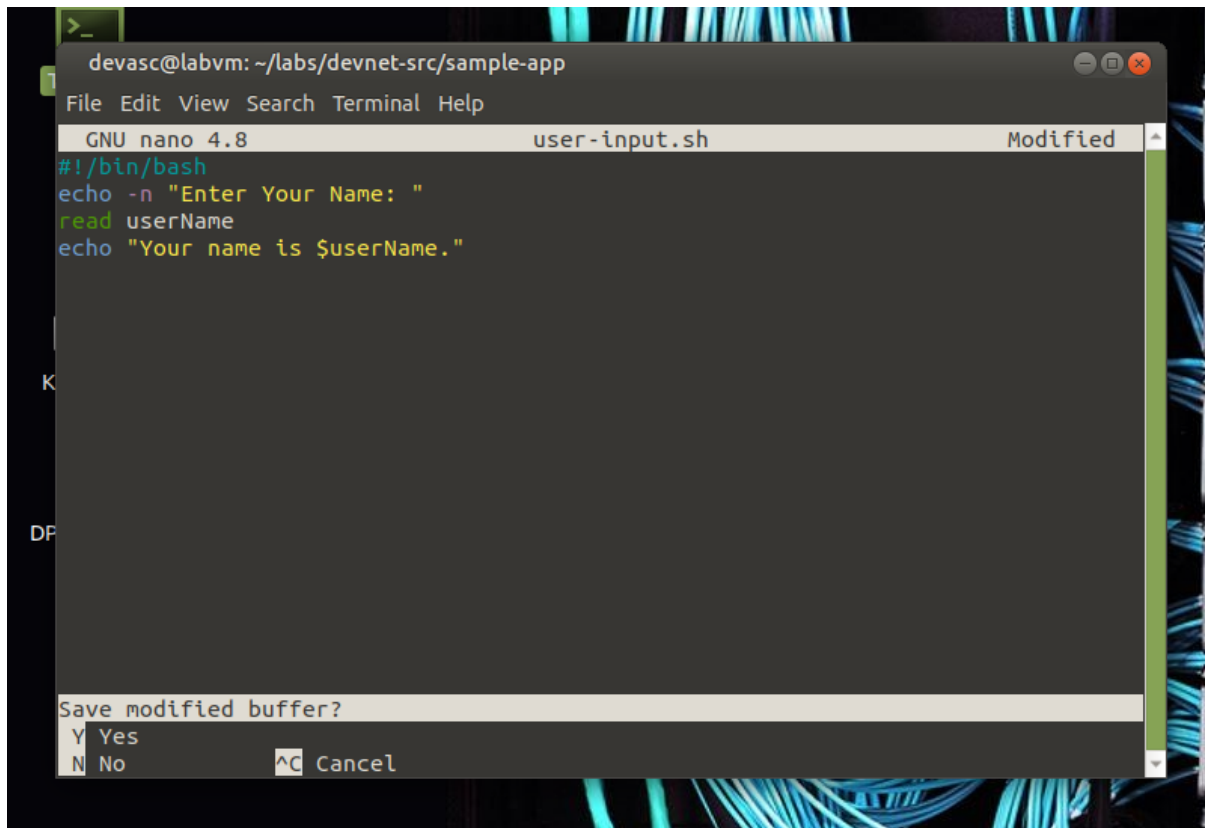
Changement de répertoire vers le dossier de travail et création du script :

```
cd ~/labs/devnet-src/sample-app
nano user-input.sh
```

Étape 4 : Ajout des commandes Bash

Le script demande le nom de l'utilisateur, le stocke dans une variable et l'affiche :

```
#!/bin/bash
echo -n "Enter Your Name: "
read userName
echo "Your name is $userName."
```



```
devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help
GNU nano 4.8 user-input.sh Modified
#!/bin/bash
echo -n "Enter Your Name: "
read userName
echo "Your name is $userName."

Save modified buffer?
Y Yes
N No ^C Cancel
```

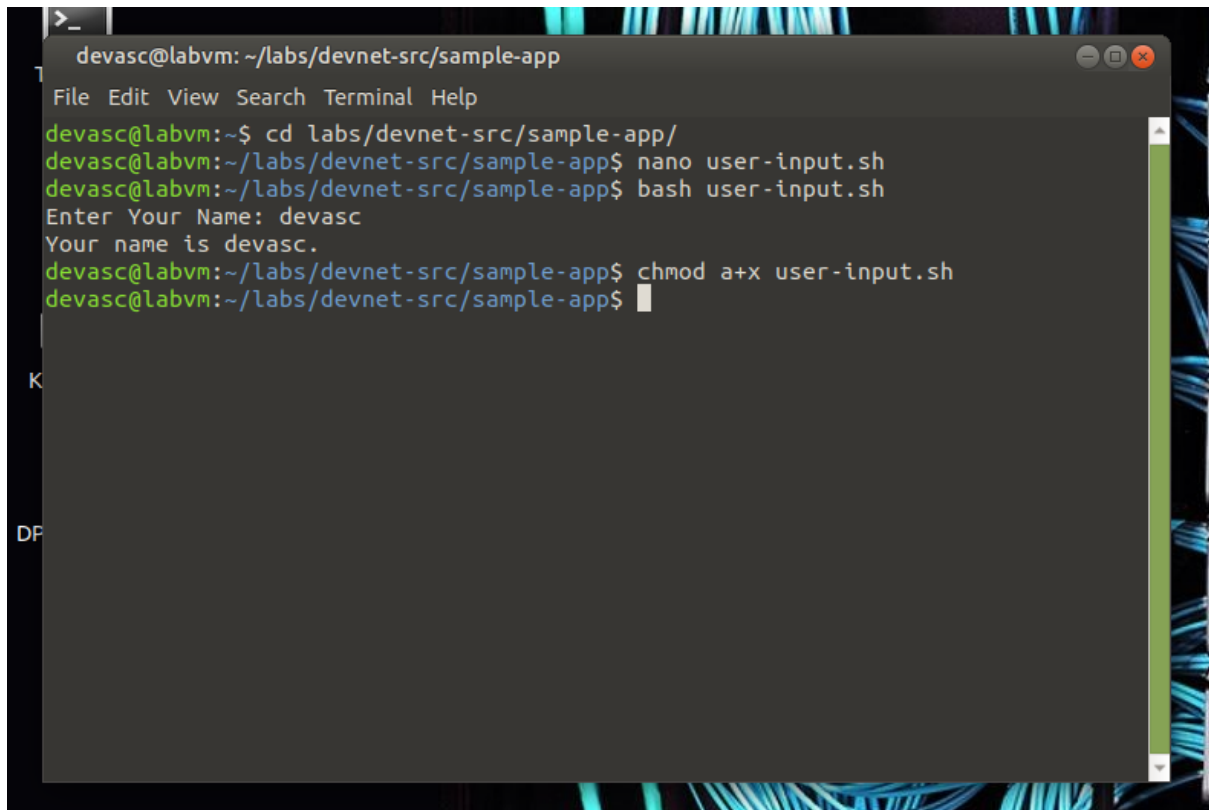
Étapes 6-7 : Exécution et permissions

Exécution directe via bash puis modification des droits d'exécution :

```
bash user-input.sh
chmod a+x user-input.sh
```

Résultat obtenu lors du test :

```
Enter Your Name: Ahmed Your name is decasc.
```

A terminal window titled 'devasc@labvm: ~/labs/devnet-src/sample-app' with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows the following commands and output:

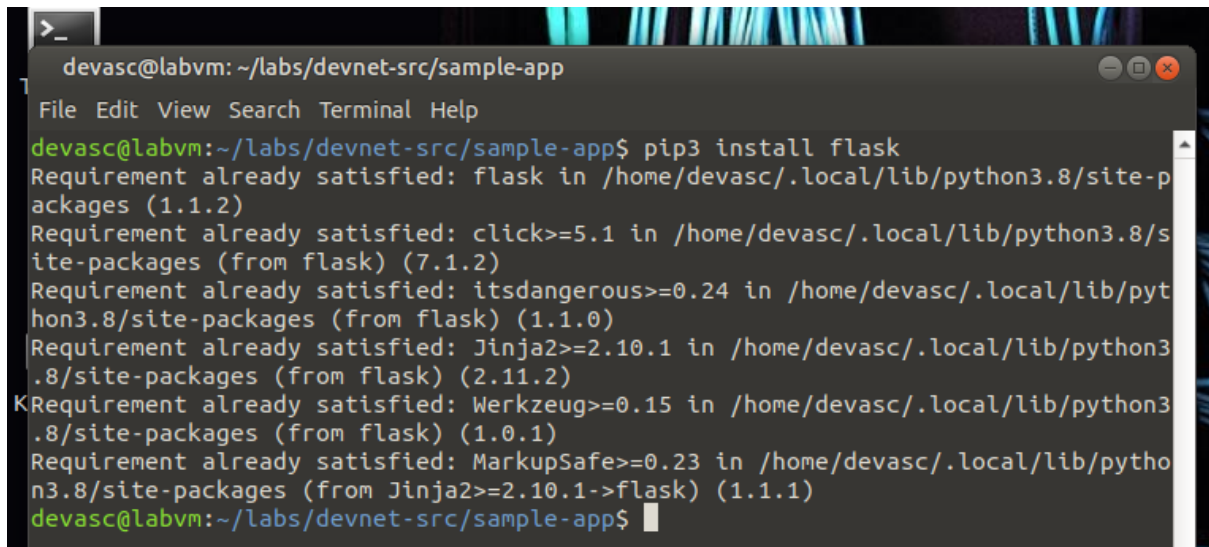
```
devasc@labvm:~$ cd labs/devnet-src/sample-app/  
devasc@labvm:~/labs/devnet-src/sample-app$ nano user-input.sh  
devasc@labvm:~/labs/devnet-src/sample-app$ bash user-input.sh  
Enter Your Name: devasc  
Your name is devasc.  
devasc@labvm:~/labs/devnet-src/sample-app$ chmod a+x user-input.sh  
devasc@labvm:~/labs/devnet-src/sample-app$
```

Partie 3 : Création de l'application Web avec Flask

Flask est un micro-framework Python pour développer des applications Web. Il permet de créer rapidement des routes HTTP et de gérer les requêtes/réponses.

Étape 1 : Installation de Flask

```
pip3 install flask
```



```
devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help
devasc@labvm:~/labs/devnet-src/sample-app$ pip3 install flask
Requirement already satisfied: flask in /home/devasc/.local/lib/python3.8/site-packages (1.1.2)
Requirement already satisfied: click>=5.1 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (7.1.2)
Requirement already satisfied: itsdangerous>=0.24 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (1.1.0)
Requirement already satisfied: Jinja2>=2.10.1 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (2.11.2)
Requirement already satisfied: Werkzeug>=0.15 in /home/devasc/.local/lib/python3.8/site-packages (from flask) (1.0.1)
Requirement already satisfied: MarkupSafe>=0.23 in /home/devasc/.local/lib/python3.8/site-packages (from Jinja2>=2.10.1->flask) (1.1.1)
devasc@labvm:~/labs/devnet-src/sample-app$
```

Étape 2-6 : Code de sample_app.py (version initiale)

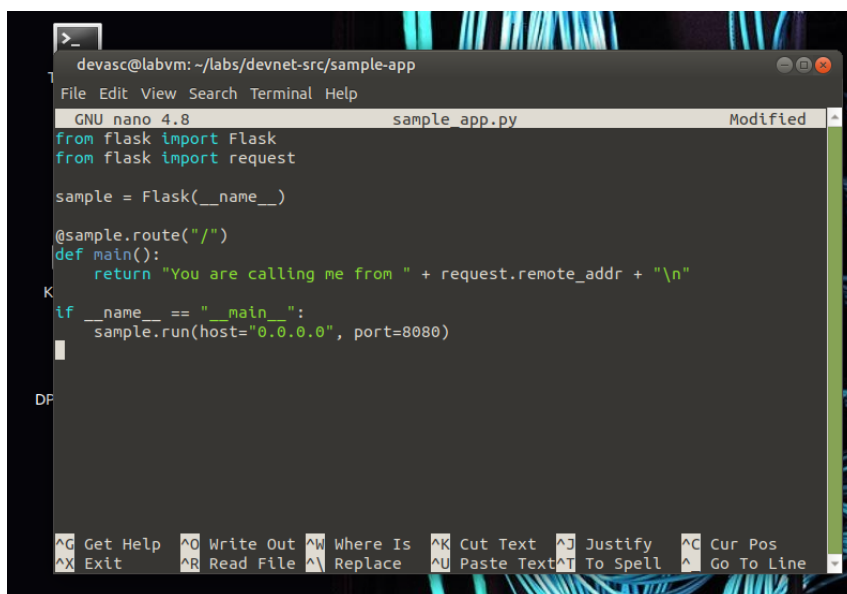
Le fichier sample_app.py est créé avec le contenu suivant :

```
from flask import Flask
from flask import request

sample = Flask(__name__)

@sample.route("/")
def main():
    return "You are calling me from " + request.remote_addr + "\n"

if __name__ == "__main__":
    sample.run(host="0.0.0.0", port=8080)
```



```
devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help
GNU nano 4.8 sample_app.py Modified
from flask import Flask
from flask import request

sample = Flask(__name__)

@sample.route("/")
def main():
    return "You are calling me from " + request.remote_addr + "\n"

if __name__ == "__main__":
    sample.run(host="0.0.0.0", port=8080)
```

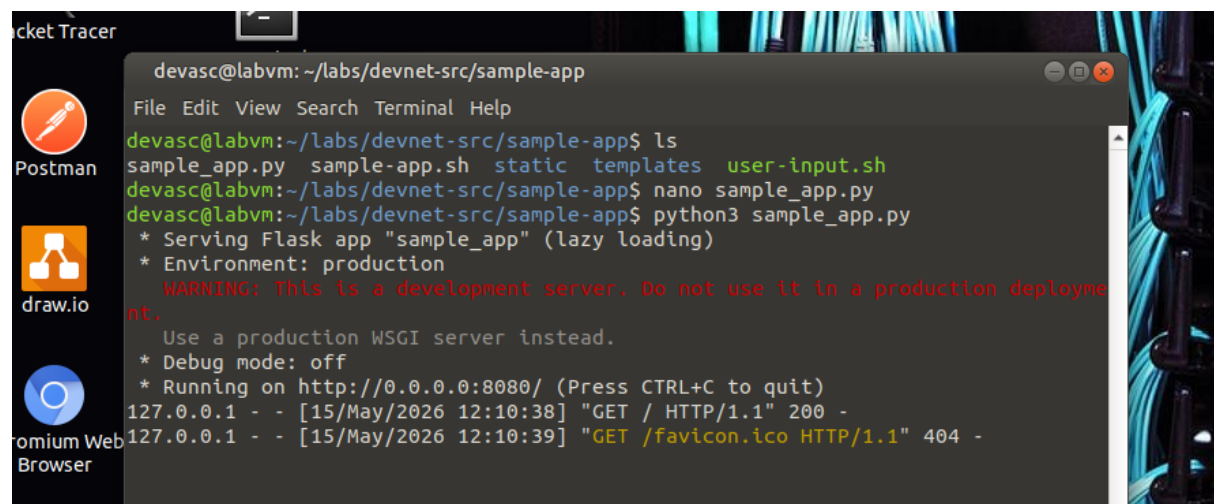
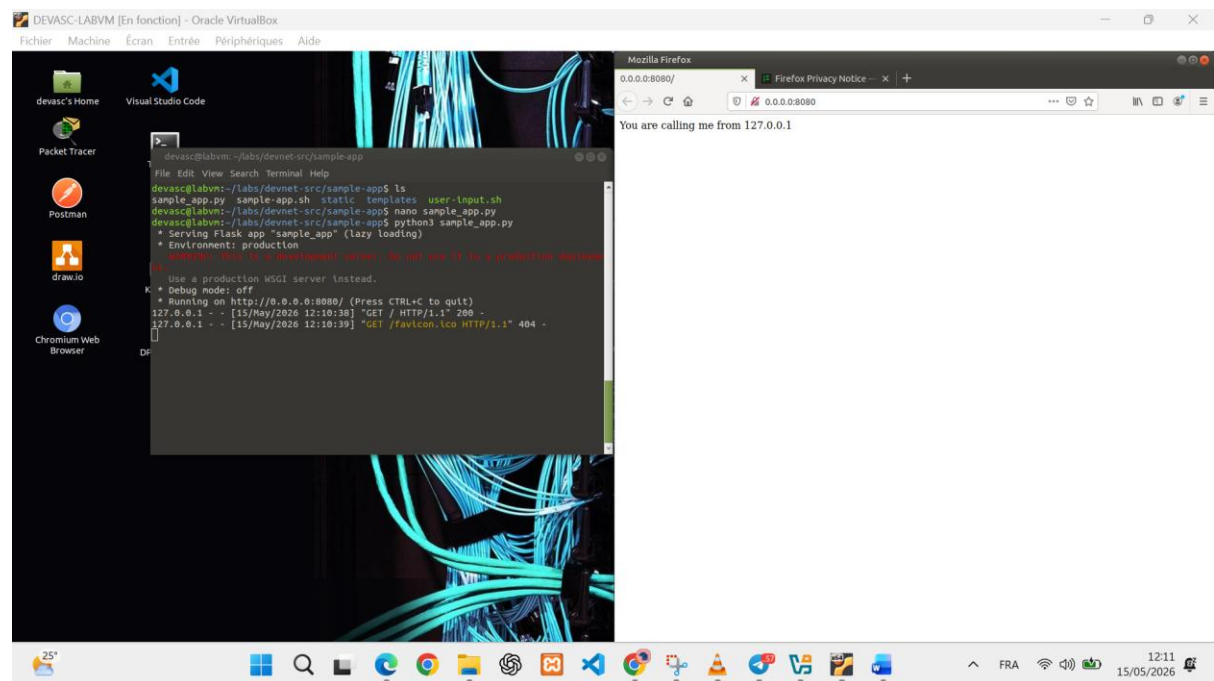
Étape 7-8 : Test du serveur

Lancement du serveur Flask :

```
python3 sample_app.py
* Serving Flask app "sample_app" * Running on http://0.0.0.0:8080/ Press CTRL+C to quit
```

Vérification avec cURL :

```
curl http://0.0.0.0:8080
You are calling me from 127.0.0.1
```



Partie 4 : Configuration avec fichiers HTML/CSS

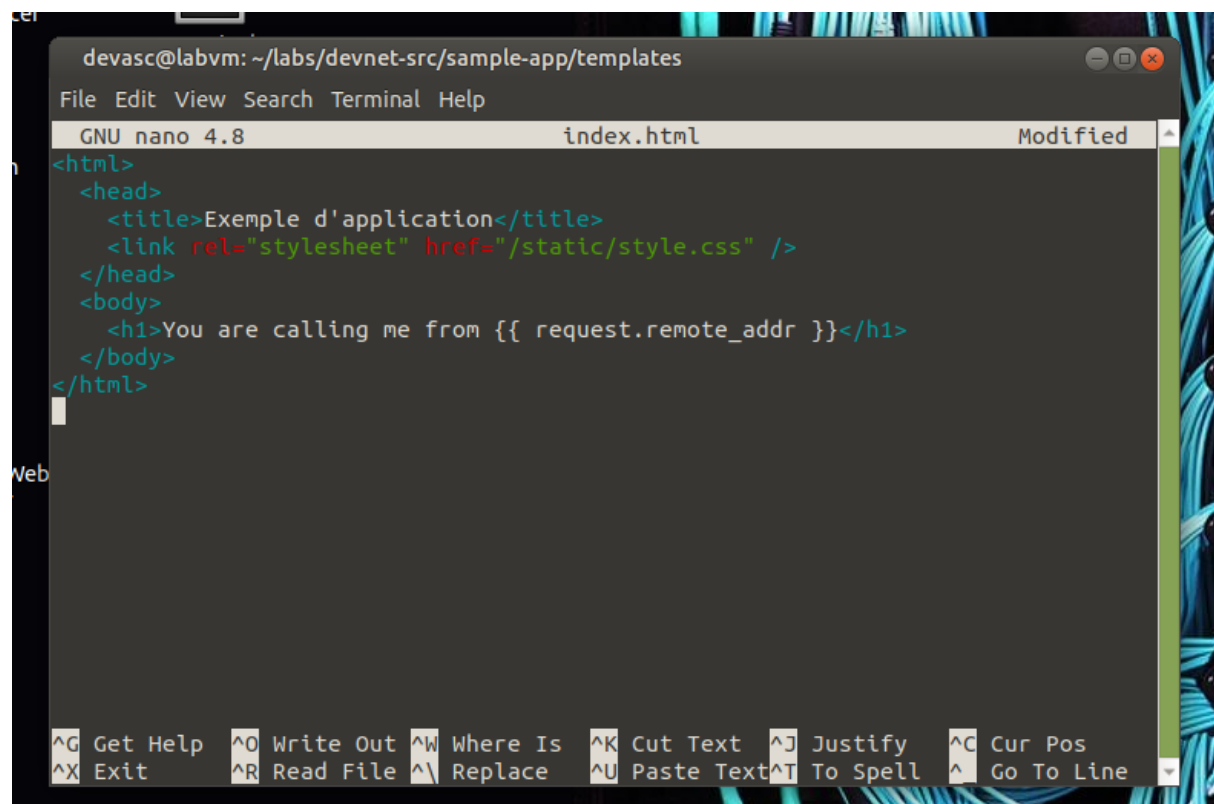
Dans cette partie, l'application est enrichie avec un fichier HTML (templates/index.html) et une feuille de style CSS (static/style.css).

Structure des répertoires

```
sample-app/  
├─ sample_app.py  
├─ templates/  
│   └─ index.html  
└─ static/  
    └─ style.css
```

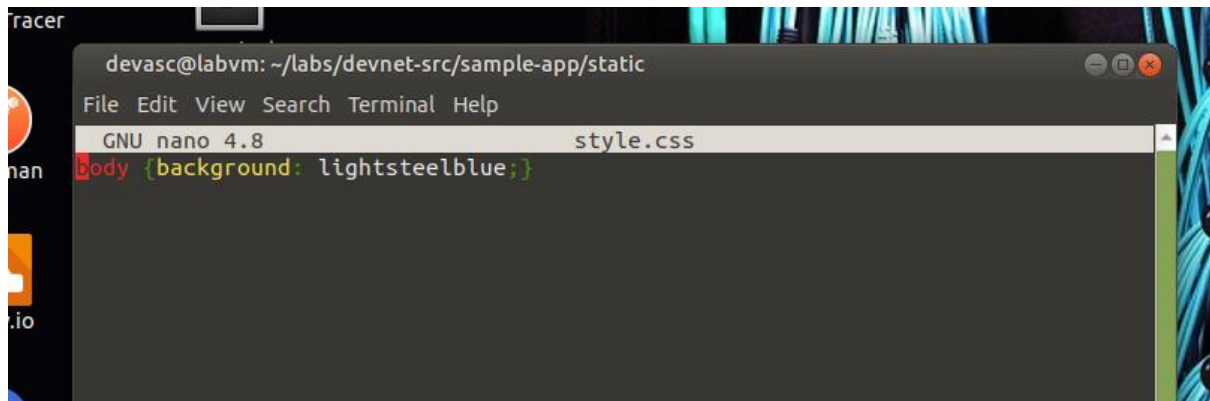
Contenu de templates/index.html

```
<html>  
  <head>  
    <title>Exemple d'application</title>  
    <link rel="stylesheet" href="/static/style.css" />  
  </head>  
  <body>  
    <h1>You are calling me from {{ request.remote_addr }}</h1>  
  </body>  
</html>
```



Contenu de static/style.css

```
body {background: lightsteelblue;}
```



Mise à jour de sample_app.py

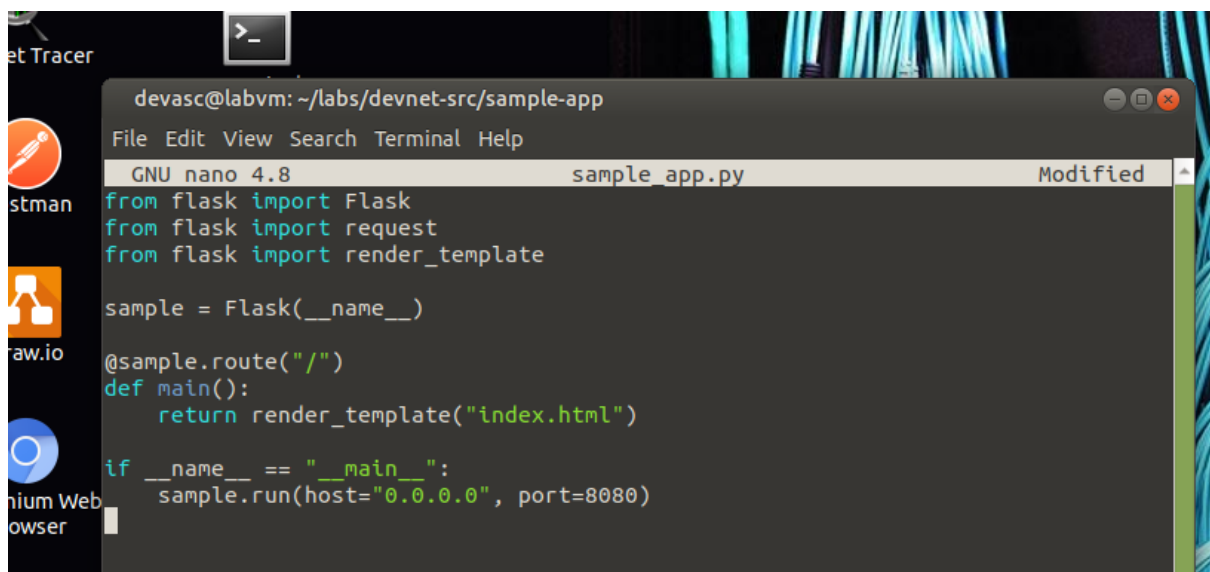
Ajout de render_template pour afficher le fichier HTML :

```
from flask import Flask
from flask import request
from flask import render_template

sample = Flask(__name__)

@sample.route("/")
def main():
    return render_template("index.html")

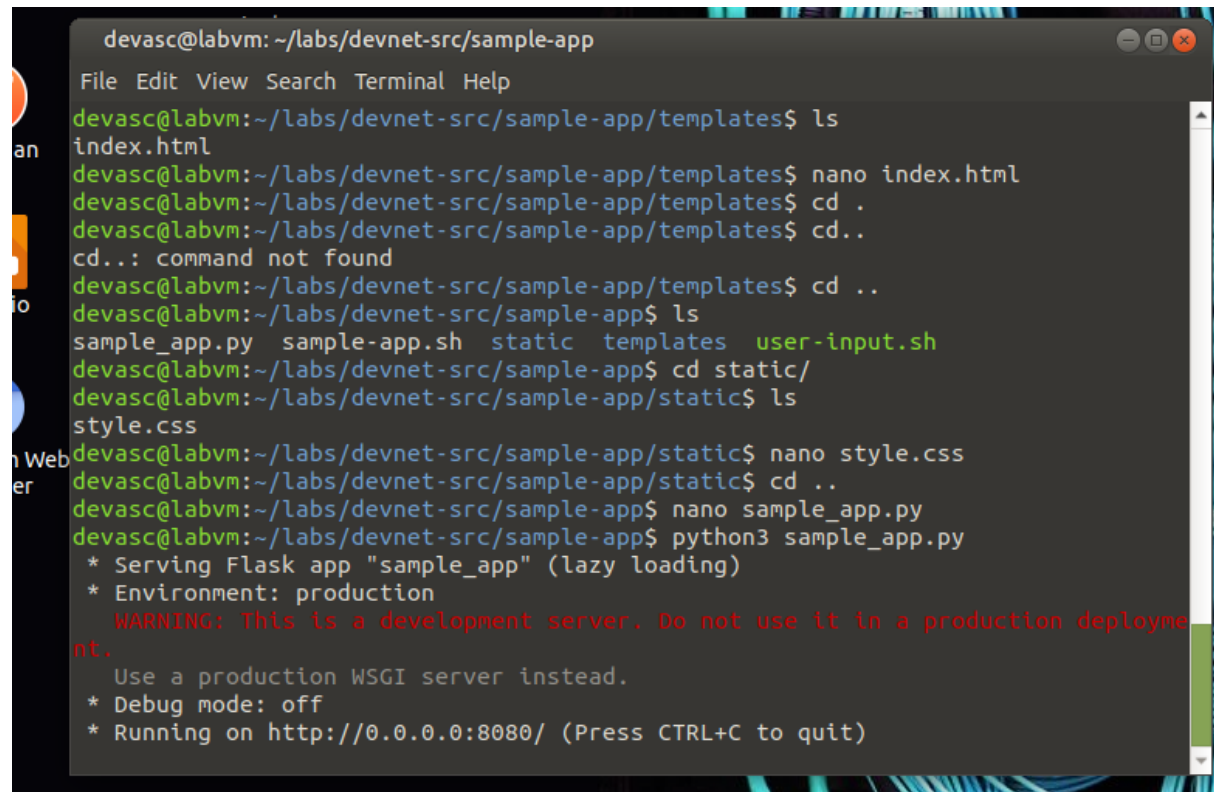
if __name__ == "__main__":
    sample.run(host="0.0.0.0", port=8080)
```



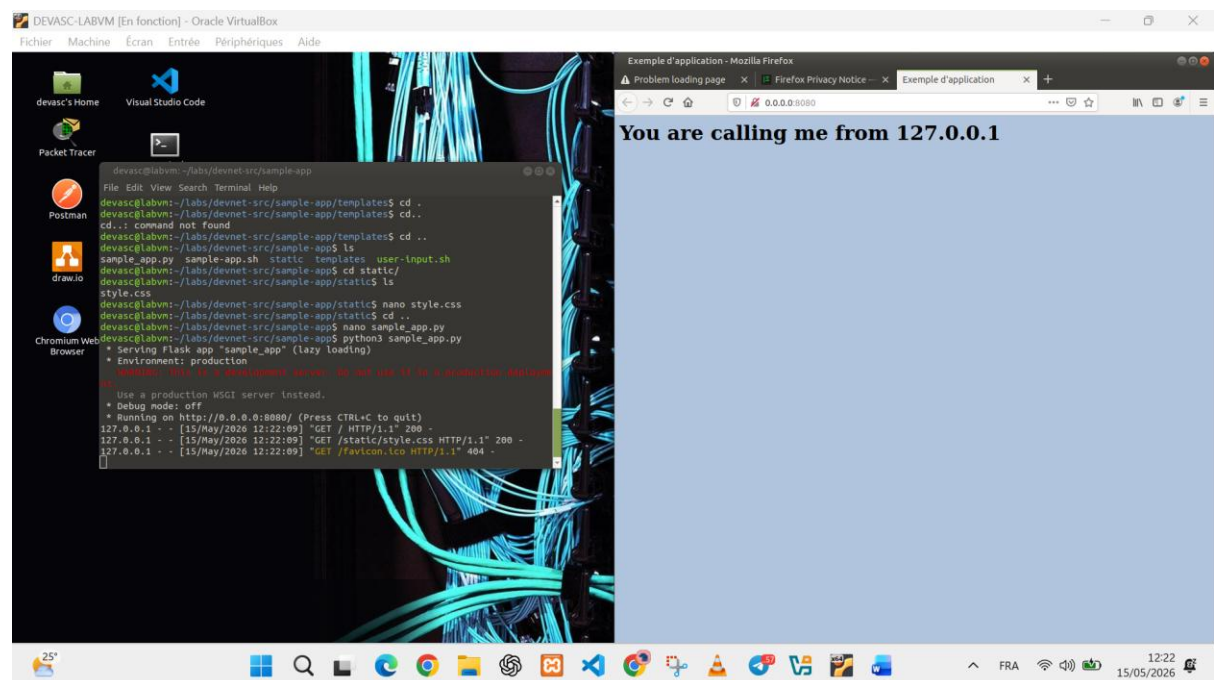
Vérification

Test avec cURL retournant le HTML complet :

```
curl http://0.0.0.0:8080
<html>  <head>      <title>Exemple d'application</title>  </head>  <body>
<h1>You are calling me from 127.0.0.1</h1>  </body> </html>
```



```
devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help
devasc@labvm:~/labs/devnet-src/sample-app/templates$ ls
index.html
devasc@labvm:~/labs/devnet-src/sample-app/templates$ nano index.html
devasc@labvm:~/labs/devnet-src/sample-app/templates$ cd .
devasc@labvm:~/labs/devnet-src/sample-app/templates$ cd..
cd..: command not found
devasc@labvm:~/labs/devnet-src/sample-app/templates$ cd ..
devasc@labvm:~/labs/devnet-src/sample-app$ ls
sample_app.py  sample-app.sh  static  templates  user-input.sh
devasc@labvm:~/labs/devnet-src/sample-app$ cd static/
devasc@labvm:~/labs/devnet-src/sample-app/static$ ls
style.css
devasc@labvm:~/labs/devnet-src/sample-app/static$ nano style.css
devasc@labvm:~/labs/devnet-src/sample-app/static$ cd ..
devasc@labvm:~/labs/devnet-src/sample-app$ nano sample_app.py
devasc@labvm:~/labs/devnet-src/sample-app$ python3 sample_app.py
* Serving Flask app "sample_app" (lazy loading)
* Environment: production
  WARNING: This is a development server. Do not use it in a production deployment.
  Use a production WSGI server instead.
* Debug mode: off
* Running on http://0.0.0.0:8080/ (Press CTRL+C to quit)
```



Partie 5 : Script Bash pour Docker

Ce script automatise entièrement la création et le lancement du conteneur Docker.

Contenu du fichier sample-app.sh

```
#!/bin/bash

# Étape 1 : Créer les répertoires temporaires
mkdir tmpdir
mkdir tmpdir/templates
mkdir tmpdir/static

# Étape 2 : Copier les fichiers
cp sample_app.py tmpdir/.
cp -r templates/* tmpdir/templates/.
cp -r static/* tmpdir/static/.

# Étape 3 : Créer le Dockerfile
echo "FROM python" > tmpdir/Dockerfile
echo "RUN pip install flask" >> tmpdir/Dockerfile
echo "COPY ./static /home/myapp/static/" >> tmpdir/Dockerfile
echo "COPY ./templates /home/myapp/templates/" >> tmpdir/Dockerfile
echo "COPY sample_app.py /home/myapp/" >> tmpdir/Dockerfile
echo "EXPOSE 8080" >> tmpdir/Dockerfile
echo "CMD python3 /home/myapp/sample_app.py" >> tmpdir/Dockerfile

# Étape 4 : Construire le conteneur Docker
cd tmpdir
docker build -t sampleapp .

# Étape 5 : Démarrer et vérifier
docker run -t -d -p 8080:8080 --name samplerunning sampleapp
docker ps -a
```

```
devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help
GNU nano 4.8 sample-app.sh Modified
#!/bin/bash

mkdir tmpdir
mkdir tmpdir/templates
mkdir tmpdir/static

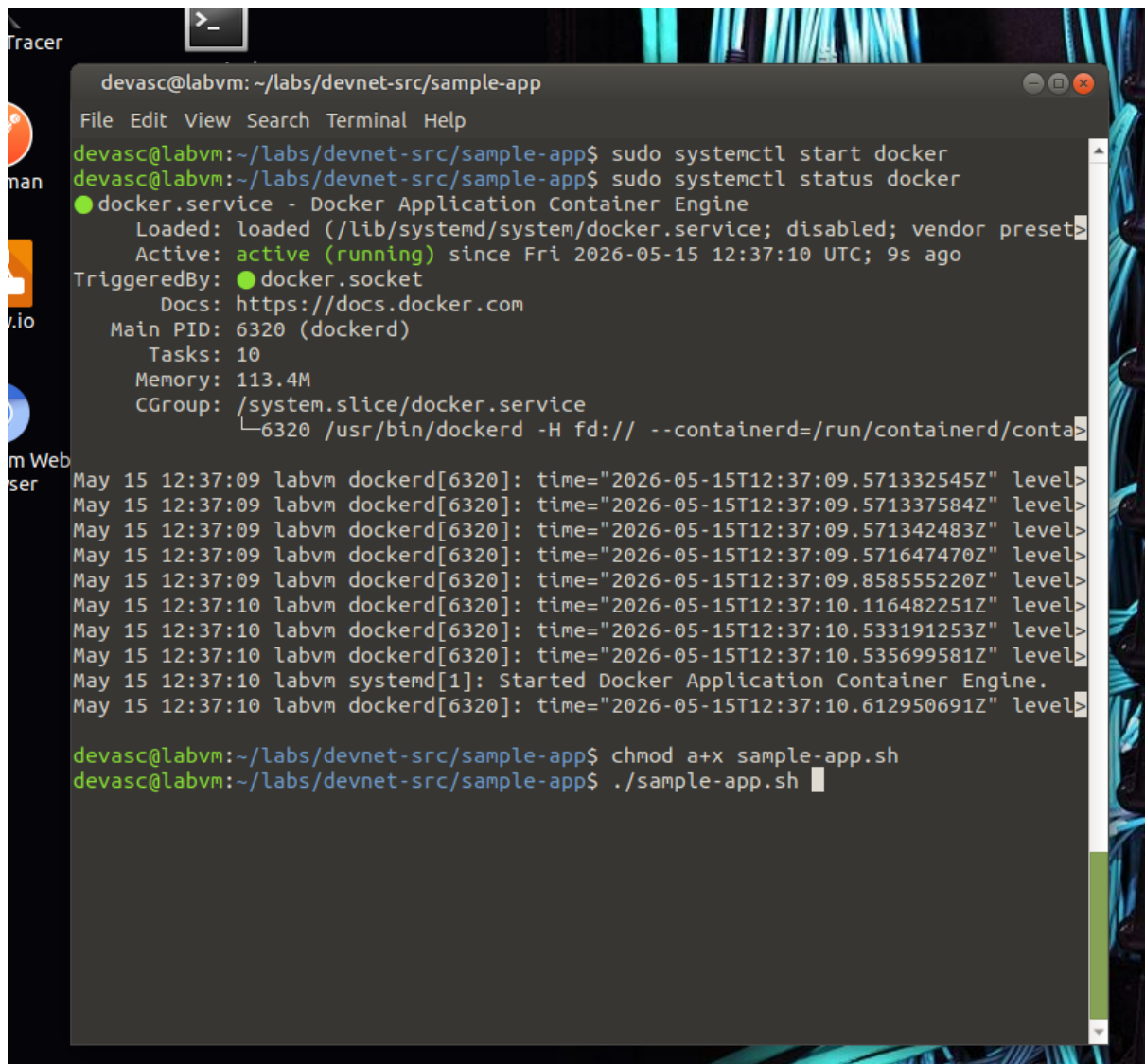
cp sample_app.py tmpdir/.
cp -r templates/* tmpdir/templates/.
cp -r static/* tmpdir/static/.

Web
echo "FROM python" > tmpdir/Dockerfile
echo "RUN pip install flask" >> tmpdir/Dockerfile
echo "COPY ./static /home/myapp/static/" >> tmpdir/Dockerfile
echo "COPY ./templates /home/myapp/templates/" >> tmpdir/Dockerfile
echo "COPY sample_app.py /home/myapp/" >> tmpdir/Dockerfile
echo "EXPOSE 8080" >> tmpdir/Dockerfile
echo "CMD python3 /home/myapp/sample_app.py" >> tmpdir/Dockerfile

cd tmpdir
docker build -t sampleapp .

docker run -t -d -p 8080:8080 --name samplerunning sampleapp
docker ps -a
```

Vérifier si docker est allumer

A terminal window titled 'devasc@labvm: ~/labs/devnet-src/sample-app' showing the command 'sudo systemctl start docker' and 'sudo systemctl status docker'. The status output shows 'docker.service - Docker Application Container Engine' is 'loaded' and 'active (running)'. Below this, a series of logs from 'dockerd[6320]' and 'systemd[1]' are visible, including timestamps and log levels. The terminal also shows the command 'chmod a+x sample-app.sh' and the execution of './sample-app.sh'.

```
devasc@labvm: ~/labs/devnet-src/sample-app
File Edit View Search Terminal Help
devasc@labvm:~/labs/devnet-src/sample-app$ sudo systemctl start docker
devasc@labvm:~/labs/devnet-src/sample-app$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/lib/systemd/system/docker.service; disabled; vendor preset>
   Active: active (running) since Fri 2026-05-15 12:37:10 UTC; 9s ago
 TriggeredBy: ● docker.socket
            Docs: https://docs.docker.com
   Main PID: 6320 (dockerd)
    Tasks: 10
   Memory: 113.4M
   CGroup: /system.slice/docker.service
           └─6320 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/conta>
May 15 12:37:09 labvm dockerd[6320]: time="2026-05-15T12:37:09.571332545Z" level>
May 15 12:37:09 labvm dockerd[6320]: time="2026-05-15T12:37:09.571337584Z" level>
May 15 12:37:09 labvm dockerd[6320]: time="2026-05-15T12:37:09.571342483Z" level>
May 15 12:37:09 labvm dockerd[6320]: time="2026-05-15T12:37:09.571647470Z" level>
May 15 12:37:09 labvm dockerd[6320]: time="2026-05-15T12:37:09.858555220Z" level>
May 15 12:37:10 labvm dockerd[6320]: time="2026-05-15T12:37:10.116482251Z" level>
May 15 12:37:10 labvm dockerd[6320]: time="2026-05-15T12:37:10.533191253Z" level>
May 15 12:37:10 labvm dockerd[6320]: time="2026-05-15T12:37:10.535699581Z" level>
May 15 12:37:10 labvm systemd[1]: Started Docker Application Container Engine.
May 15 12:37:10 labvm dockerd[6320]: time="2026-05-15T12:37:10.612950691Z" level>

devasc@labvm:~/labs/devnet-src/sample-app$ chmod a+x sample-app.sh
devasc@labvm:~/labs/devnet-src/sample-app$ ./sample-app.sh
```

Le Dockerfile généré

Voici le contenu du fichier tempdir/Dockerfile créé par le script :

```
FROM python
RUN pip install flask
COPY ./static /home/myapp/static/
COPY ./templates /home/myapp/templates/
COPY sample_app.py /home/myapp/
EXPOSE 8080
CMD python3 /home/myapp/sample_app.py
```

```
devasc@labvm: ~/labs/devnet-src/sample-app/tempdir
File Edit View Search Terminal Help
GNU nano 4.8 Dockerfile
FROM python
RUN pip install flask
COPY ./static /home/myapp/static/
COPY ./templates /home/myapp/templates/
COPY sample_app.py /home/myapp/
EXPOSE 8080
CMD python3 /home/myapp/sample_app.py

[ Read 7 lines ]
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Line
```

Partie 6 : Build, exécution et vérification

Je n'ai pas pu exécuter le script et télécharger tous les fichiers par problème de connexion

